

**Mohammed Reabai J 250701435-2025-CSE** ▾**M2****Started on** Monday, 24 August 2026, 7:45 PM**State** Finished**Completed on** Monday, 24 August 2026, 8:02 PM**Time taken** 17 mins 36 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2  int main(){
3      int num;
4      int count=0;
5      int i;
6      scanf("%d",&num);
7      for(i=1;i<=num;++i){
8          count++;
9          if(num%i==0){
10             count++;
11         }
12         count++;
13     }
14     count++;
15     printf("%d",count);
16     return 0;
17 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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